

Lab 4 – VibeCoding: Chat History

Patruni SreeVishnu Pavan · PES1UG24AM187 · AIML-D

Repo: SETAPESU26/06_coin_collector · Tool: Claude (Claude Code, model Claude Opus 5.5) · Exported 08/10/2026

Shown: every message from the start of Lab 4, plus each tool call Claude made (shell commands, file edits) and a short excerpt of its output.

STUDENT

[screenshot attached: repository link]

STUDENT

@ "C:\Users\HP OMEN\Downloads\Lab_4_VibeCoding_Student_handout.pdf"
https://github.com/SETAPESU26/06_coin_collector

❏ **Bash** – Extract text from the Lab 4 handout

```
PYTHONIOENCODING=utf-8 python -I -c " import pymupdf,sys d=pymupdf.open(sys.argv[1]); print(len(d))
for i,p in enumerate(d): print('====PAGE',i+1); print(p.get_text()) " "C:/Users/HP
OMEN/Downloads/Lab_4_VibeCoding_Student_handout.pdf"
```

Exit code 1

Traceback (most recent call last):

File "<string>", line 4, in <module>

```
for i,p in enumerate(d): print('====PAGE',i+1); print(p.get_text())
```

~~~~~^

File "C:\Users\HP OMEN\AppData\Local\Python\pythoncore-3.14-64\Lib\encodings\cp1252.py", line 19, in encode

```
return codecs.charmap_encode(input,self.errors,encoding_table)[0]
```

~~~~~^

UnicodeEncodeE ...

❏ **Bash** – Extract Lab 4 handout text with UTF-8 output

```
python -I -c " import pymupdf,sys sys.stdout.reconfigure(encoding='utf-8') d=pymupdf.open(sys.argv[1])
for i,p in enumerate(d): print('====PAGE',i+1); print(p.get_text()) " "C:/Users/HP
OMEN/Downloads/Lab_4_VibeCoding_Student_handout.pdf"
```

====PAGE 1

Lab 4: VibeCoding

Objective

Use Vibe Coding tools to fix the broken code and add features to the same. Learn how

to prompt to get the job done in under three to four attempts. Clone a repo assigned

to you, make changes and push to your repo in your personal account.

THIS IS AN INDIVIDUAL ASSIGNMENT. WORK INDIVIDUALLY!!

Duration

60 minutes

Software Requirements

- Access to Vibe Coding tool. (ChatGPT, Gemini, Claude etc)

Instructions:

1. Go to the r ...

❏ **PowerShell** – Check git, gh auth, pygame and ffmpeg availability

```
git --version; gh --version 2>$null | Select-Object -First 1; gh auth status 2>&1 | Select-Object -First 4; git config --global user.name; git config --global user.email; python -c "import pygame; print(pygame.version.ver)" 2>&1 | Select-Object -Last 1; ffmpeg -version 2>$null | Select-Object -First ...
```

Exit code 1

```
git version 2.53.0.windows.2
gh version 2.102.0 (2026-09-30)
```

github.com

✓ Logged in to github.com account vishnu23-sys (keyring)

- Active account: true

- Git operations protocol: https

Vishnu Patruni

patruni.vishnu@gmail.com

ModuleNotFoundError: No module named 'pygame'

ffmpeg : The term 'ffmpeg' is not recognized as the name of a cmdlet, function, script file, or operable program.

Check the spelling of the name, or if a path was included, verify that the path is correc ...

❏ **PowerShell** – List user's GitHub repos and view the assigned repo

```
gh repo list vishnu23-sys --limit 50 --json name,visibility,updatedAt | ConvertFrom-Json | Format-Table -AutoSize; gh repo view SETAPESU26/06_coin_collector --json name,description,defaultBranchRef | Out-String
```

| name | updatedAt | visibility |
|----------------------|----------------------|------------|
| ----- | ----- | ----- |
| PES1UG24AM187-SE-LAB | 2026-10-08T16:33:42Z | PUBLIC |
| FlowPilot | 2026-06-04T11:55:11Z | PUBLIC |
| PESU-SGPA-Calculator | 2026-05-28T07:57:56Z | PUBLIC |
| Quiz-System-TDP | 2026-05-28T05:57:48Z | PUBLIC |

```
{"defaultBranchRef":{"name":"main"},"description":"","name":"06_coin_collector"}
```

CLAUDE

Your lab repo looks like `vishnu23-sys/PES1UG24AM187-SE-LAB`. I'll clone it and the assigned game repo to look at both.

❏ **PowerShell** – Clone assigned repo and personal lab repo, list contents

```
New-Item -ItemType Directory -Force "D:\Vishnu\sem5\se\lab4" | Out-Null Set-Location
"D:\Vishnu\sem5\se\lab4" git clone https://github.com/SETAPESU26/06_coin_collector.git 2>&1 | Select-
Object -Last 1 git clone https://github.com/vishnu23-sys/PES1UG24AM187-SE-LAB.git 2>&1 | Select-Object
-Last 1 Get ...
```

```
git : Cloning into '06_coin_collector'...
```

```
At line:3 char:1
```

```
+ git clone https://github.com/SETAPESU26/06_coin_collector.git 2>&1 | ...
```

```
+ ~~~~~
```

```
+ CategoryInfo          : NotSpecified: (Cloning into '06_coin_collector'...:String) [], RemoteException
```

```
+ FullyQualifiedErrorId : NativeCommandError
```

```
git : Cloning into 'PES1UG24AM187-SE-LAB'...
```

```
At line:4 char:1
```

```
+ git clone https://github.com/vishnu23-sys/PES1UG24AM187-SE-LAB. ...
```

❏ **Bash** – Read the coin collector README and source files

```
cd "/d/Vishnu/sem5/se/lab4/06_coin_collector" && cat README.md && for f in coin-collector/main.py
coin-collector/requirements.txt coin-collector/game/*.py; do echo "==== $f"; cat -n "$f"; done
```

```
# Coin Collector Lab
```

This project is a single-topic top-down Coin Collector game using

****Pygame****. It introduces students to per-frame collision bookkeeping,

entity variety, an obstacle/lives system, and round timing, using a

small, readable object-oriented codebase.

```
---
```

```
## What's Provided
```

A working Coin Collector game with:

- A player that moves around a play area with the arrow keys
- Coins scattered around the play area that award points on contact

- A running score displa ...

❖ **Bash** – Install pygame and imageio-ffmpeg

```
pip install --quiet pygame imageio imageio-ffmpeg 2>&1 | grep -v notice; python -I -c "import pygame, imageio_ffmpeg; print(pygame.version.ver, imageio_ffmpeg.get_ffmpeg_exe())"
```

Exit code 1

error: subprocess-exited-with-error

Getting requirements to build wheel did not run successfully.

exit code: 1

[76 lines of output]

Skipping Cython compilation

WARNING, No "Setup" File Exists, Running "buildconfig/config.py"

Using WINDOWS MSYS2 configuration...

Installing pre-built dependencies

Installing mingw-w64-x86_64-SDL2

For help with compilation see:

<https://www.pygame.org/wiki/CompileWindows>

To contribute to pygame developme ...

CLAUDE

pygame has no ready-made build for your Python 3.14. I'll install `pygame-ce` instead: it's the maintained drop-in replacement, imported as `pygame` the same way.

❖ **Bash** – Install pygame-ce and imageio-ffmpeg

```
pip install --quiet pygame-ce imageio imageio-ffmpeg 2>&1 | grep -v notice; python -I -c "import pygame, imageio_ffmpeg; print(pygame.version.ver, imageio_ffmpeg.get_ffmpeg_exe())"
```

WARNING: The scripts f2py.exe and numpy-config.exe are installed in 'C:\Users\HP OMEN\AppData\Local\Python\pythoncore-3.14-64\Scripts' which is not on PATH.

Consider adding this directory to PATH or, if you prefer to suppress this warning, use `--no-warn-script-location`.

WARNING: The scripts imageio_download_bin.exe and imageio_remove_bin.exe are installed in 'C:\Users\HP OMEN\AppData\Local\Python\pythoncore-3.14-64\Scripts' which is not on PATH.

Consider adding this directory to PATH or, i ...

CLAUDE

pygame-ce installed. Next I'm copying the starter game into a `Lab-4` folder in your lab repo and committing it untouched.

❖ **Bash** – Copy starter game into Lab-4 folder of lab repo

```
cd "/d/Vishnu/sem5/se/lab4" && L=PES1UG24AM187-SE-LAB/Lab-4 && mkdir -p "$L/videos" && cp -r 06_coin_collector/coin-collector "$L/" && cp 06_coin_collector/README.md "$L/coin-collector/README.md" && cp 06_coin_collector/.gitignore "$L/coin-collector/.gitignore" && cat "$L/coin-collector/.gitignore" ...
```

__pycache__/
__pycache__/

*.pyc

.venv/
__pycache__/

venv/?? Lab-4/
98681b0 Add files via upload

```
bfe8447 Add files via upload
5b0945c Add files via upload
5925070 Add initial README for Lab 1 project
main
Shell cwd was reset to C:\Users\HP OMEN\AppData\Roaming\Claude\scratch-workspaces\9229de7a-8d92-42e8-a871-c09651b003bd\774eab02-ac9d-4a7f-be5c-0227ee4d2369\scratch-2026-10-08-66172f
```

CLAUDE

Now the recorder: it runs the real game with scripted arrow-key input and saves the frames as a 10-second MP4.

✧ Write

```
C:\Users\HPOMEN~1\AppData\Local\Temp\claude\C--Users-HP-OMEN-AppData-Roaming-Claude-scratch-workspaces-9229de7a-8d92-42e8-a871-c09651b003bd-774eab02-ac9d-4a7f-be5c-0227ee4d2369-scratch-2026-10-08-66172f\89a27c05-21c4-4a7a-8b5a-c3cf85b54af8\scratchpad\record_demo.py
```

File created successfully at: C:\Users\HPOMEN~1\AppData\Local\Temp\claude\C--Users-HP-OMEN-AppData-Roaming-Claude-scratch-workspaces-9229de7a-8d92-42e8-a871-c09651b003bd-774eab02-ac9d-4a7f-be5c-0227ee4d2369-scratch-2026-10-08-66172f\89a27c05-21c4-4a7a-8b5a-c3cf85b54af8\scratchpad\record_demo.py (file state is current in your context – no need to Read it back)

✧ Bash – Record before-fix gameplay video and grab sample frames

```
SP="/c/Users/HPOMEN~1/AppData/Local/Temp/claude/C--Users-HP-OMEN-AppData-Roaming-Claude-scratch-workspaces-9229de7a-8d92-42e8-a871-c09651b003bd-774eab02-ac9d-4a7f-be5c-0227ee4d2369-scratch-2026-10-08-66172f/89a27c05-21c4-4a7a-8b5a-c3cf85b54af8/scratchpad" L="/d/Vishnu/sem5/se/lab4/PES1UG24AM187-SE-L ...
```

wrote 300 frames (10.0s), final score 167
10.0

We had to kill ffmpeg to stop it.

Shell cwd was reset to C:\Users\HP OMEN\AppData\Roaming\Claude\scratch-workspaces\9229de7a-8d92-42e8-a871-c09651b003bd\774eab02-ac9d-4a7f-be5c-0227ee4d2369\scratch-2026-10-08-66172f

✧ Read

```
C:\Users\HP OMEN\AppData\Local\Temp\lab2\before_150.png
```

CLAUDE

The "before" video shows the bug clearly: the score reached 167 from walking over about 6 coins. Committing the untouched starter code and the before video separately:

✧ Bash – Commit starter code and before video

```
cd "/d/Vishnu/sem5/se/lab4/PES1UG24AM187-SE-LAB" && git add Lab-4/coin-collector && git commit -q -m "Lab-4: add Coin Collector starter code (SETAPESU26/06_coin_collector) Co-Authored-By: Claude Opus 5.5 <noreply@anthropic.com>" && git add Lab-4/videos/before.mp4 && git commit -q -m "Lab-4: add 10s ...
```

```
da409ff Lab-4: add 10s gameplay video before changes (coin bug visible)
2efbf0f Lab-4: add Coin Collector starter code (SETAPESU26/06_coin_collector)
98681b0 Add files via upload
Shell cwd was reset to C:\Users\HP OMEN\AppData\Roaming\Claude\scratch-workspaces\9229de7a-8d92-42e8-a871-c09651b003bd\774eab02-ac9d-4a7f-be5c-0227ee4d2369\scratch-2026-10-08-66172f
```

CLAUDE

Task 1: fixing the coin bug. Each collected coin is now removed and a fresh one spawns, so it scores exactly once.

✧ Edit

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\game_engine.py

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

✧ Edit

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\game_engine.py

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

✧ Read

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\game_engine.py

```
1      """
2      GameEngine: owns the player and all coins.
3
4      Starter version: one coin type, no obstacles, no timer yet. Coin
5      collection also has a known bug (see how `update` uses check_collection
6      below) that Task 1 asks you to fix - collected coins are never removed,
7      so standing on one keeps awarding points every frame.
8      """
9
10     import random
11     import pygame
12
13     from game.player import Player
14     from game.coin import Coin
15     from game.collection import check_collection
16     from game.ren ...
```

✧ Edit

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\game_engine.py

✧ Edit

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\game_engine.py

The file D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\game_engine.py has been updated successfully. (file state is current in your context – no need to Read it back)

The file D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\game_engine.py has been updated successfully. (file state is current in your context – no need to Read it back)

CLAUDE

Checking the fix: I'll stand the player on a coin for 2 seconds of game frames and confirm it scores once.

✧ Bash – Verify Task 1 fix headlessly and commit

```
cd "/d/Vishnu/sem5/se/lab4/PES1UG24AM187-SE-LAB/Lab-4/coin-collector" && SDL_VIDEODRIVER=dummy python
-c " import random; random.seed(1) from game.game_engine import GameEngine e = GameEngine(); c =
e.coins[0] e.player.x, e.player.y = c.x, c.y for _ in range(120): e.update() print('score after stand
...

```

score after standing on one coin for 120 frames: 1 | coins on field: 6 | coin still present: False

814dd44 Task 1: collect each coin exactly once

Shell cwd was reset to C:\Users\HP OMEN\AppData\Roaming\Claude\scratch-workspaces\9229de7a-8d92-42e8-a871-c09651b003bd\774eab02-ac9d-4a7f-be5c-0227ee4d2369\scratch-2026-10-08-66172f

CLAUDE

Task 1 is fixed and committed. Task 2: bronze, silver and gold coins (1, 3 and 5 points), each with its own colour and size.

✧ Write

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\coin.py

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

✧ Read

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\renderer.py

```
1      """
2      renderer: all pygame drawing lives here, kept separate from game logic.
3      """
4
5      import pygame
6
7      WIDTH, HEIGHT = 700, 500
8      WINDOW_SIZE = (WIDTH, HEIGHT)
9
10     COLOR_BG = (35, 45, 35)
11     COLOR_PLAYER = (80, 180, 255)
12     COLOR_TEXT = (255, 255, 255)
13
14
15     def draw_scene(surface, player, coins):
16         surface.fill(COLOR_BG)
17         for coin in coins:
18             pygame.draw.circle(surface, coin.color, (int(coin.x), int(coin.y)), coin.radius)
19             pygame.draw.rect(surface, COLOR_ ...
```

✧ Read

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\coin.py

```
1      """
2      Coin: a static collectible circle. Drawn as a circle, hit-tested as a
3      bounding square around it.
4      """
5
6      import pygame
7
8
9      class Coin:
10         def __init__(self, x, y, radius=12, value=1, color=(230, 190, 60)):
11             self.x = x
12             self.y = y
13             self.radius = radius
14             self.value = value
15             self.color = color
16
17         def get_rect(self):
18             return pygame.Rect(
19                 int(self.x - self.radius), int(self.y - self.radius),
20                 ...
```

✧ Write

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\coin.py

✧ Edit

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\renderer.py

The file D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\coin.py has been updated successfully. (file state is current in your context – no need to Read it back)

The file D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\renderer.py has been updated successfully. (file state is current in your context – no need to Read it back)

CLAUDE

Now the engine side of Task 2:

✎ Edit

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\game_engine.py

✎ Edit

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\game_engine.py

✎ Edit

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\game_engine.py

The file D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\game_engine.py has been updated successfully. (file state is current in your context – no need to Read it back)

The file D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\game_engine.py has been updated successfully. (file state is current in your context – no need to Read it back)

The file D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\game_engine.py has been updated successfully. (file state is current in your context – no need to Read it back)

✎ Edit

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\game_engine.py

✎ Bash – Test coin type values and render a frame

```
cd "/d/Vishnu/sem5/se/lab4/PES1UG24AM187-SE-LAB/Lab-4/coin-collector" && SDL_VIDEODRIVER=dummy python  
-c " import random, pygame; random.seed(3) from game.game_engine import GameEngine from game.coin  
import Coin pygame.init(); s = pygame.display.set_mode((700,500)); f = pygame.font.SysFont('consolas ...
```

The file D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\game_engine.py has been updated successfully. (file state is current in your context – no need to Read it back)

```
bronze awarded 1 expected 1  
silver awarded 3 expected 3  
gold awarded 5 expected 5  
Counter({'bronze': 617, 'silver': 299, 'gold': 84})  
Shell cwd was reset to C:\Users\HP OMEN\AppData\Roaming\Claude\scratch-workspaces\9229de7a-8d92-42e8-a871-  
c09651b003bd\774eab02-ac9d-4a7f-be5c-0227ee4d2369\scratch-2026-10-08-66172f
```

✎ Read

C:\Users\HP OMEN\AppData\Local\Temp\lab2\task2.png

✎ Bash – Commit Task 2

```
cd "/d/Vishnu/sem5/se/lab4/PES1UG24AM187-SE-LAB" && git add -A Lab-4/coin-collector && git commit -q -  
m "Task 2: add bronze, silver and gold coin types Bronze = 1, silver = 3, gold = 5 points. Each type  
has its own colour and size, rarer coins spawn less often, and a legend at the bottom of the scr ...
```

warning: in the working copy of 'Lab-4/coin-collector/game/coin.py', LF will be replaced by CRLF the next time Git touches it

23b3151 Task 2: add bronze, silver and gold coin types

Shell cwd was reset to C:\Users\HP OMEN\AppData\Roaming\Claude\scratch-workspaces\9229de7a-8d92-42e8-a871-c09651b003bd\774eab02-ac9d-4a7f-be5c-0227ee4d2369\scratch-2026-10-08-66172f

✎ Write

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\obstacle.py

✎ Write

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\collection.py

File created successfully at: D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\obstacle.py
(file state is current in your context – no need to Read it back)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

🔗 Read

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\collection.py

```
1      """
2      collection: figures out which coins the player has collected this frame.
3      """
4
5
6      def check_collection(player, coins):
7          """
8          Returns the list of coins the player is currently overlapping.
9          """
10         player_rect = player.get_rect()
11         collected = []
12         for coin in coins:
13             if player_rect.colliderect(coin.get_rect()):
14                 collected.append(coin)
15         return collected
16
```

CLAUDE

Still on Task 3: adding obstacle collision detection.

🔗 Edit

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\collection.py

🔗 Edit

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\collection.py

🔗 Read

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\game_engine.py

The file D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\collection.py has been updated successfully. (file state is current in your context – no need to Read it back)

The file D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\collection.py has been updated successfully. (file state is current in your context – no need to Read it back)

```
1      """
2      GameEngine: owns the player and all coins.
3
4      Bronze, silver and gold coins (see game/coin.py), no obstacles, no timer
5      yet. Each coin is collected exactly once: `update` removes it as soon as
6      it is scored and spawns a new coin somewhere else.
7      """
8
9      import random
10     import pygame
11
12     from game.player import Player
13     from game.coin import Coin, COIN_TYPES, random_coin_type
14     from game.collection import check_collection
15     from game.renderer import WIDTH, HEIGHT
16
17     NUM_C ...
```

🔗 Write

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\game_engine.py

🔗 Edit

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\renderer.py

Edit

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\renderer.py

The file D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\game_engine.py has been updated successfully. (file state is current in your context – no need to Read it back)

The file D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\renderer.py has been updated successfully. (file state is current in your context – no need to Read it back)

The file D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\renderer.py has been updated successfully. (file state is current in your context – no need to Read it back)

CLAUDE

Testing Task 3: lives lost per touch, obstacles staying inside the play area, and the game stopping at 0 lives.

Bash – Test obstacle bounds, lives and invulnerability

```
cd "/d/Vishnu/sem5/se/lab4/PES1UG24AM187-SE-LAB/Lab-4/coin-collector" && SDL_VIDEODRIVER=dummy python -c " import random, pygame; random.seed(5) from game.game_engine import GameEngine from game.renderer import WIDTH, HEIGHT pygame.init(); s = pygame.display.set_mode((700,500)); f = pygame.font.SysF ...
```

```
obstacles always inside play area: True
lives lost at frames: [0, 91, 182] | lives: 0 | game_over: True
lives after a 10-frame touch: 2
Shell cwd was reset to C:\Users\HP OMEN\AppData\Roaming\Claude\scratch-workspaces\9229de7a-8d92-42e8-a871-c09651b003bd\774eab02-ac9d-4a7f-be5c-0227ee4d2369\scratch-2026-10-08-66172f
```

Read

C:\Users\HP OMEN\AppData\Local\Temp\lab2\task3.png

CLAUDE

Works as intended. One small issue: when the game ends mid-blink, the player can stay invisible. Fixing that, then committing.

Edit

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\game_engine.py

Bash – Commit Task 3

```
cd "/d/Vishnu/sem5/se/lab4/PES1UG24AM187-SE-LAB" && git add -A Lab-4/coin-collector && git commit -q -m "Task 3: add moving obstacles and a 3-life system Three red obstacles drift around and bounce off the edges of the play area. Touching one costs a life, then the player blinks and is invulnerable ...
```

The file D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\game_engine.py has been updated successfully. (file state is current in your context – no need to Read it back)

```
warning: in the working copy of 'Lab-4/coin-collector/game/game_engine.py', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'Lab-4/coin-collector/game/obstacle.py', LF will be replaced by CRLF the next time Git touches it
168637e Task 3: add moving obstacles and a 3-life system
Shell cwd was reset to C:\Users\HP OMEN\AppData\Roaming\Claude\scratch-workspaces\9229de7a-8d92-42e8-a871-c09651b003bd\774eab02-ac9d-4a7f-be5c-0227ee4d2369\scratch-2026-10-08-66172 ...
```

CLAUDE

Task 3 is committed. Task 4: a 30-second round timer, a clear end-of-round screen, and R to start a new round.

🔗 Read

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\main.py

🔗 Read

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\game_engine.py

```
1      """
2      Coin Collector (Lab Starter)
3
4      Run with: python3 main.py
5
6      Controls: Arrow keys to move.
7      """
8
9      import pygame
10
11     from game.game_engine import GameEngine
12     from game.renderer import WINDOW_SIZE
13
14
15     def main():
16         pygame.init()
17         screen = pygame.display.set_mode(WINDOW_SIZE)
18         pygame.display.set_caption("Coin Collector")
19         clock = pygame.time.Clock()
20         font = pygame.font.SysFont("consolas", 22)
21
22         engine = GameEngine()
23         running = ...
24
25
26
27
28
29     class GameEngine:
30         def __init__(self):
31             self.player = Player(x=WIDTH / 2, y=HEIGHT / 2)
32             self.obstacles = [self._random_obstacle() for _ in range(NUM_OBSTACLES)]
33             self.coins = [self._random_coin() for _ in range(NUM_COINS)]
34             self.score = 0
35             self.lives = START_LIVES
36             self.invulnerable = 0
37             self.game_over = False
38
39         def _random_coin(self):
```

🔗 Edit

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\main.py

🔗 Edit

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\main.py

The file D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\main.py has been updated successfully.
(file state is current in your context – no need to Read it back)

The file D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\main.py has been updated successfully.
(file state is current in your context – no need to Read it back)

CLAUDE

Now the engine changes: round state, countdown, end reason and reset.

🔗 Bash – Add round timer, end reason and reset to the game engine

```
cd "/d/Vishnu/sem5/se/lab4/PES1UG24AM187-SE-LAB/Lab-4/coin-collector/game" && python -I - <<'EOF' p =
'game_engine.py' s = open(p, encoding='utf-8').read() R = [ (''Bronze, silver and gold coins (see
game/coin.py) and moving obstacles, no timer yet. Each coin is collected exactly once: `update` rem ...
```

ok

Shell cwd was reset to C:\Users\HP OMEN\AppData\Roaming\Claude\scratch-workspaces\9229de7a-8d92-42e8-a871-c09651b003bd\774eab02-ac9d-4a7f-be5c-0227ee4d2369\scratch-2026-10-08-66172f

🔗 Read

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\rendererer.py

```
55         x += 24 + label.get_width() + 18
56
57
58     def draw_text(surface, font, text, pos, color=COLOR_TEXT):
59         surface.blit(font.render(text, True, color), pos)
60
61
62     def draw_banner(surface, font, text):
63         surf = font.render(text, True, (255, 220, 80))
64         rect = surf.get_rect(center=(surface.get_width() // 2, surface.get_height() // 2))
65         surface.blit(surf, rect)
66
```

🔗 Edit

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\rendererer.py

🔗 Edit

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\rendererer.py

🔗 Edit

D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\rendererer.py

The file D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\rendererer.py has been updated successfully.

The file D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\rendererer.py has been updated successfully.

The file D:\Vishnu\sem5\se\lab4\PES1UG24AM187-SE-LAB\Lab-4\coin-collector\game\rendererer.py has been updated successfully.

CLAUDE

Testing Task 4: the round ends at 30 seconds or at 0 lives, and R resets everything.

🔗 Bash – Test timer, round end, restart and lives end

```
cd "/d/Vishnu/sem5/se/lab4/PES1UG24AM187-SE-LAB/Lab-4/coin-collector" && SDL_VIDEODRIVER=dummy python
-c " import random, pygame; random.seed(11) from game.game_engine import GameEngine pygame.init(); s =
pygame.display.set_mode((700,500)); f = pygame.font.SysFont('consolas', 22) R = pygame.event.Ev ...
```

```
timer round ended after 30.02 s | reason: Time's up! | time_left: 0.0
frozen after round end (time_left unchanged): True
other keys ignored: True
after R -> round_over: False score: 0 lives: 3 time_left: 30.0
lives round ended | reason: Out of lives! | time_left: 26.95
Shell cwd was reset to C:\Users\HP OMEN\AppData\Roaming\Claude\scratch-workspaces\9229de7a-8d92-42e8-a871-
c09651b003bd\774eab02-ac9d-4a7f-be5c-0227ee4d2369\scratch-2026-10-08-66172f
```

🔗 Read

C:\Users\HP OMEN\AppData\Local\Temp\lab2\task4_over.png

🔗 Read

C:\Users\HP OMEN\AppData\Local\Temp\lab2\task4_play.png

CLAUDE

Both end conditions work and R resets the round. Last check: running the real `main.py` loop for 1.5 seconds to make sure it starts and quits cleanly.

🔗 Bash – Smoke-test `main.py` and commit Task 4

```
cd "/d/Vishnu/sem5/se/lab4/PES1UG24AM187-SE-LAB/Lab-4/coin-collector" && SDL_VIDEODRIVER=dummy timeout 20 python -c " import pygame, main orig = pygame.display.set_mode def sm(*a, **k): r = orig(*a, **k); pygame.time.set_timer(pygame.QUIT, 1500); return r pygame.display.set_mode = sm main.main() ...
```

main loop ran and quit cleanly

ac8597c Task 4: add a 30-second timed round with restart

Shell cwd was reset to C:\Users\HP OMEN\AppData\Roaming\Claude\scratch-workspaces\9229de7a-8d92-42e8-a871-c09651b003bd\774eab02-ac9d-4a7f-be5c-0227ee4d2369\scratch-2026-10-08-66172f

CLAUDE

All 4 tasks are committed. Recording the "after" video now.

🔗 Bash – Record after-changes gameplay video and sample frames

```
SP="/c/Users/HPOMEN~1/AppData/Local/Temp/claude/C--Users-HP-OMEN-AppData-Roaming-Claude-scratch-workspaces-9229de7a-8d92-42e8-a871-c09651b003bd-774eab02-ac9d-4a7f-be5c-0227ee4d2369-scratch-2026-10-08-66172f/89a27c05-21c4-4a7a-8b5a-c3cf85b54af8/scratchpad" L="/d/Vishnu/sem5/se/lab4/PES1UG24AM187-SE-L ...
```

wrote 384 frames (12.8s), final score 4

frames 384 duration 12.8

Shell cwd was reset to C:\Users\HP OMEN\AppData\Roaming\Claude\scratch-workspaces\9229de7a-8d92-42e8-a871-c09651b003bd\774eab02-ac9d-4a7f-be5c-0227ee4d2369\scratch-2026-10-08-66172f

🔗 Bash – Combine sampled after-video frames into one sheet

```
cd "$LOCALAPPDATA/Temp/lab2" && python -I -c " from PIL import Image ims=[Image.open(f'after_{i}.png') for i in (45,120,170,210,250,374)] W,H=ims[0].size; sheet=Image.new('RGB',(W*3,H*2)) for k,im in enumerate(ims): sheet.paste(im,((k%3)*W,(k//3)*H)) sheet.resize((W*3//2,H)).save('after_sheet.png')"
```

Shell cwd was reset to C:\Users\HP OMEN\AppData\Roaming\Claude\scratch-workspaces\9229de7a-8d92-42e8-a871-c09651b003bd\774eab02-ac9d-4a7f-be5c-0227ee4d2369\scratch-2026-10-08-66172f

🔗 Read

C:\Users\HP OMEN\AppData\Local\Temp\lab2\after_sheet.png

🔗 Bash – Tune autopilot timing and re-record the after video

```
SP="/c/Users/HPOMEN~1/AppData/Local/Temp/claude/C--Users-HP-OMEN-AppData-Roaming-Claude-scratch-workspaces-9229de7a-8d92-42e8-a871-c09651b003bd-774eab02-ac9d-4a7f-be5c-0227ee4d2369-scratch-2026-10-08-66172f/89a27c05-21c4-4a7a-8b5a-c3cf85b54af8/scratchpad" cd "$SP" && python -I - <<'EOF' s = open('re ...
```

```
97:         restart_at = t + 1.6
```

```
100:         restart_at, end_at = float("inf"), max(t + 1.4, 10.0)
```

```
101:         elif t < 3.0 or end_at is not None:
```

wrote 300 frames (10.0s), final score 4

frames 300 duration 10.0

```
[1.0, 3.0, 4.7, 6.0, 7.2, 8.3, 9.0, 9.9]
```

Shell cwd was reset to C:\Users\HP OMEN\AppData\Roaming\Claude\scratch-workspaces\9229de7a-8d92-42e8-a871-c09651b003bd\774eab02-ac9d-4a7f-be5c-0227ee4d2369\scratch-2026-10-08-66172f

🔗 Read

C:\Users\HP OMEN\AppData\Local\Temp\lab2\after_sheet.png

CLAUDE

The after video is exactly 10 seconds. It shows coins being collected, lives being lost, the "Out of lives / Final score" screen, then R starting a fresh round. Committing it, then the chat history.

🔗 Bash – Commit after video

```
cd "/d/Vishnu/sem5/se/lab4/PES1UG24AM187-SE-LAB" && git add Lab-4/videos/after.mp4 && git commit -q -m "Lab-4: add 10s gameplay video after changes Shows coins collected once with bronze/silver/gold values, obstacles costing lives, the round-over screen with the final score, and R starting a fresh ..."
```

🔗 ToolSearch

```
select:mcp_ccd_session_mgmt__export_transcript
```

353ffb7 Lab-4: add 10s gameplay video after changes

Shell cwd was reset to C:\Users\HP OMEN\AppData\Roaming\Claude\scratch-workspaces\9229de7a-8d92-42e8-a871-c09651b003bd\774eab02-ac9d-4a7f-be5c-0227ee4d2369\scratch-2026-10-08-66172f